

The Future of International Environmental Law

Edited by **David Leary** and **Balakrishna Pisupati**

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1

Introduction

David Leary and Balakrishna Pisupati

International environmental law: Responding to a world of problems

In its *Global Environment Outlook (GEO 4)* published in 2007 the United Nations Environment Programme (UNEP) paints a bleak picture of the state of the global environment and the connection of environmental degradation with development and human well-being. In the introductory words to this report, UNEP observes:

Imagine a world in which environmental change threatens people's health, physical security, material needs and social cohesion. This is a world beset by increasingly intense and frequent storms, and by rising sea levels. Some people experience extensive flooding, while others endure intense droughts. Species extinction occurs at rates never before witnessed. Safe water is increasingly limited, hindering economic activity. Land degradation endangers the lives of millions of people. This is the world today.¹

Climate change is one of the most significant of the many global environmental challenges. While our understanding of the causes and impacts of climate change were slow to emerge, it is no understatement to say that climate change is the greatest threat facing humanity and all life on Earth today. While all regions of the world will be impacted by climate change, it is the poor and especially the developing countries that are the most vulnerable. A detailed examination of the projected impacts of climate change are beyond the scope of this chapter, but a few examples

drawn from the *Fourth Assessment Report* of the Intergovernmental Panel on Climate Change (IPCC) are worth noting to highlight the nature and scale of the challenges posed by climate change. For example, the IPCC has noted that by 2020, in some parts of Africa, “yields from rain-fed agriculture could be reduced by up to 50%”.² This in turn is projected to compromise agricultural production and further adversely affect food security and exacerbate malnutrition in Africa.³ In Asia, coastal areas, especially the heavily populated megadelta regions in South, East and Southeast Asia, are projected to be at increased risk of flooding.⁴ For small island communities, sea level rise is expected to exacerbate inundation, storm surge, erosion and other coastal hazards and threaten vital infrastructure, settlements and facilities upon which island communities rely.⁵

Climate change will not only affect humans and human communities. By 2020, significant loss of biodiversity is projected to occur in many of the world’s ecologically rich sites such as the Great Barrier Reef in Australia.⁶ There is also “a risk of significant biodiversity loss through species extinction in many areas of tropical Latin America”.⁷ While the loss of biodiversity is already occurring, and in part is caused by climate change, it is also the result of many other human impacts. Threats to biodiversity come from varied sources including habitat destruction such as deforestation and modification and destruction of ecosystems for industrial, agricultural and other activities, overexploitation of resources (through activities such as overfishing and hunting), pollution, introduced species, to mention but a few.

The loss of biodiversity is occurring at an alarming rate. An update of the *IUCN Red List* published in 2008, the world’s leading source of information on the status of biodiversity, makes for sober reading. This update presented an assessment of only 44 838 of the several millions of species thought to exist on Earth. Key figures from this assessment worth highlighting include:

- There were 869 recorded extinctions, with 804 species listed as extinct and 65 listed as extinct in the wild;
- The number of extinctions increases to 1159 if the 290 critically endangered species tagged as “possibly extinct” are included;
- 16 928 species are threatened with extinction (3246 are critically endangered, 4770 are endangered and 8912 are vulnerable);
- 3796 species are listed as near threatened; and
- 5570 species have insufficient information to determine their threat status and are listed as data deficient.⁸

Climate change and the alarming rates of biodiversity loss are but two examples of the many environmental problems humanity has faced over the past few decades. These challenges are in turn closely linked to devel-

opment and human well-being. The global community has taken steps to seek to tackle many of these environmental challenges. These attempts have seen a veritable bloom in the development of international environmental law. There are now more than 900 multilateral and over 1500 bilateral treaties and other international agreements dealing with environmental issues.⁹ These are in addition to the thousands of “soft law” instruments such as declarations and plans of action which have been the product of environmental diplomacy in recent decades. But despite the proliferation of international environmental agreements, environmental degradation has proceeded and new environmental challenges have continued to emerge.

But given the experience, some might say the failure, of international environmental law to deal with existing environmental challenges before turning to consider new and emerging challenges (the subject matter of much of this book), it is useful to consider the history of international environmental law and what lessons we may learn from that history. With that in mind this chapter seeks to set the scene for subsequent chapters by briefly sketching the history of international environmental law and to suggest some lessons that might help shape approaches to new and emerging issues in international environmental law, such as those discussed in this book. The chapter will then conclude with an outline of the structure and content of the rest of this book.

The development of international environmental law: The convergence of environment and development

International environmental law has arguably now gone through four key phases in its development. While treaties dealing with access to fishery resources, for example, can be identified at least as far back as 1351 with the conclusion of a fishery treaty between England and Castile, it was not until the mid-nineteenth century that the first series of treaties dealing with what we would regard today as environmental issues began to emerge. These included a range of fishery treaties that sought to guard against overexploitation of certain fish stocks; the first treaty dealing specifically with the protection of migratory birds, the first regional wildlife protection convention and conventions dealing with polluted waterways.¹⁰

But the most significant development of this first phase of international environmental law was the emergence of key principles from two international arbitrations, both involving the United States. The first of these was the 1893 Pacific Fur Seal Arbitration¹¹ which involved a dispute between the United States and the United Kingdom in relation to the

protection of fur seals in the Bering Sea from overexploitation. This case rejected any suggestion that states had the right to assert jurisdiction by enacting measures relating to the conservation of living resources outside their jurisdiction, even if that meant the extinction of the species.¹² Any such measure requires the consent of all states concerned.¹³ The other major development during this first phase of the development of international environmental law was the emergence in the Trail Smelter Arbitration of the principle that “no state has the right to use or permit the use of its territory in such a manner as to cause injury by fumes in or to the territory of another or the properties or persons therein”.¹⁴ The Trail Smelter Arbitration is considered the pivotal case in the emergence of the duty to prevent transboundary environmental harm, which is now largely accepted as a principle of customary international law.

However, the emergence of international environmental law through decisions of arbitral tribunals and other international judicial bodies has been the exception rather than the rule. For the large part, international environmental law has emerged through bilateral and multilateral negotiation of treaties and “soft law” instruments such as declarations and plans of action.

During the first phase of the development of international environmental law most of the treaties that were developed were purely utilitarian in character; efforts at protecting or conserving specific species were motivated largely by their usefulness rather than environmental protection per se.¹⁵ But by the late 1960s, a period that arguably represents the emergence of a second phase of the development of international environmental law, serious concerns were being expressed about growing evidence of environmental degradation and it was realized that this degradation was closely linked to unsustainable levels of economic development and population growth. A number of studies during this period predicted a very bleak future for humanity if these trends continued unabated. For example, the Club of Rome in its seminal study *The Limits to Growth*, concluded:

If the present growth trends in world population, industrialization, pollution, food production, and resource depletion continue unchanged, the limits to growth on this planet will be reached sometime within the next one hundred years. The most probable result will be a rather sudden and uncontrollable decline in both population and industrial capacity.¹⁶

By the time of the lead-up to the United Nations Conference on the Human Environment (UNCHE) in Stockholm in 1972, an understanding was emerging that development and environmental issues were clearly interdependent. As early as 1962 the United Nations General Assembly,

in Resolution 1831 (XVII), noted that “to be effective measures to preserve natural resources, flora and fauna should be taken at the earliest possible moment simultaneously with economic development, including industrialization, and urbanization”.¹⁷

Likewise in the run-up to the Stockholm conference the *Founex Report* commented:

... concern for environment must not and need not detract from the commitment of the world community – developing and more industrialised alike – to the overriding task of development of the developing regions of the world ... the environment problem has to be placed in its proper perspective both in the developed and the developing countries. It should be treated as a problem of the most *efficient synthesis* of developmental and environmental concerns at different stages of social transitions ... it must be emphasized in all international forums, including the Stockholm Conference, that it is for the developed countries to reassure the developing world that their growing environmental concern will not hurt the continued development of the developing world.¹⁸

How to bring about the “efficient synthesis” of developmental and environmental concerns, as suggested by the *Founex Report*, has been a key theme in debate on global environmental issues in the subsequent forty years, especially at pivotal diplomatic conferences such as the 1972 UNCHE in Stockholm, the 1992 United Nations Conference on Environment and Development (UNCED) in Rio de Janeiro and the World Summit on Sustainable Development (WSSD) in Johannesburg in 2002. Woven through key documents emerging from these meetings, including such “soft law” declarations as the Stockholm Declaration,¹⁹ Agenda 21,²⁰ the Rio Declaration²¹ and the Johannesburg Declaration,²² and numerous treaties negotiated since Stockholm, the right to development and protection of the environment were increasingly linked.

The Stockholm Conference marked the beginning of the third and modern era of international environmental law. The Stockholm meeting produced a number of significant outcomes including the establishment of UNEP and the adoption of the Stockholm Declaration²³ setting out twenty-six Principles and an Action Plan containing 109 recommendations.²⁴ As Philippe Sands has observed:

The Stockholm Conference set the scene for international activities at the regional and global level, and influenced legal and institutional developments up to and beyond UNCED. Developments in this period are of two types: those directly related to Stockholm and follow-up actions; and those indirectly related thereto. The period was marked by: a proliferation of international environmental organisations (including those established by treaty) and greater

efforts by existing institutions to address environmental issues; the development of new sources of international environmental obligations from acts of such organisations; new environmental norms established by treaty; the development of new techniques for implementing environmental standards; including environmental impact assessment and access to information; and the formal integration of environment and development, particularly in relation to international trade and development assistance.²⁵

By the 1980s it was firmly recognized that measures aimed at environmental protection had to also take account of the need for development. In particular the “efficient synthesis” of developmental and environmental concerns was given major impetus by the 1987 report of the World Commission on Environment and Development, chaired by Norwegian Prime Minister Gro Harlem Brundtland (the so-called Brundtland Report).²⁶ The Brundtland Report characterized this as the need to ensure “sustainable development”, that is, “development that meets the needs of the present without compromising the ability of future generations to meet their own needs”.²⁷ The Brundtland Report led in turn to the convening of the UNCED (the Earth Summit) in Rio de Janeiro in Brazil in 1992. The Earth Summit adopted three non-binding instruments: the Rio Declaration,²⁸ the UNCED Forest Principles²⁹ and Agenda 21. In addition the Convention on Biological Diversity³⁰ and the UN Framework Convention on Climate Change³¹ were opened for signature at the Earth Summit.

Following the Earth Summit it was clear that international environmental law had come of age. Today it is possible to discern the existence of several key principles of international environmental law including significant new principles such as the precautionary principle, the “polluter pays” principle and the principle of common but differentiated responsibility.³² A number of these principles, such as the principle of common but differentiated responsibility, as well as the idea of sustainable development reflected the synthesis of environmental and development concerns.

Has international environmental law failed?

Despite the proliferation of international environmental agreements, environmental degradation has continued and new environmental challenges have continued to emerge. Why is it that, despite the huge volume of international environmental law, the state of the global environment has continued to worsen? There are many complex reasons for this and it is possible to identify a number of key points that have been put forward.

Some argue that it is the sheer volume of international legal instruments or treaty congestion that is partly to blame. As Edith Brown Weiss has observed:

The number and variety of environmental agreements has reached the point that some critics ask whether they may not severely strain the physical and organizational capacity of countries to handle them. There are signs of treaty congestion, in the form of separate negotiating fora, separate secretariats and funding mechanisms, overlapping provisions and inconsistencies between agreements, and severe demands on local capacity to participate in negotiations, meetings of parties, and associated activities. This affects the international community as a whole, since there will always be limited resources to address difficult issues and some countries may suffer particular inequities in their ability to participate effectively in new regimes ... [w]ith such a large number of international agreements, there is great potential for overlapping provisions in agreements, inconsistencies in obligations, significant gaps in coverage, and duplication of goals and responsibilities ... [I]nternational environmental law has developed in a piecemeal, almost random, manner ... [T]reaty congestion also contributes to significant inefficiencies in implementing international agreements. There are usually separate secretariats, monitoring processes, scientific councils, financing mechanisms, technical assistance programs and dispute resolution procedures ... [F]inally, treaty congestion leads to overload at the national level in negotiating and implementing the agreements ... Even industrialised states with well-developed regulatory mechanisms and bureaucracies show signs of being overwhelmed. As attention shifts to the need to comply with existing agreements, the burden on the administrative capacity of states will become more acute.³³

Clearly therefore, as new environmental challenges arise, careful thought needs to be given as to whether new law or institutions are needed, or whether it might be more appropriate for existing laws, mechanisms and institutions to be utilized or adapted to meet new environmental challenges.

Part of the explanation for failure of international environmental law to address global environmental challenges also lies in the fact that, throughout the development of international environmental law, little attention was paid to the “effectiveness” of this new body of law. The debate on the effectiveness of international environmental law more broadly, or regimes for specific issues, has received considerable attention in both the academic and policy literature in recent years. By far the most comprehensive and detailed examination of this concept so far is that provided by Chambers.³⁴ After reviewing numerous studies by legal scholars and social scientists (and international relations literature in particular) he characterizes effectiveness of international environmental law in the following terms:

... there are three critical elements to the measurement of effectiveness that we can be certain of establishing with the knowledge that we have now. The first is a measurement of a treaty's effectiveness based on its performance data and compared with its objectives. A second determinant of effectiveness, which has not been widely argued in the literature thus far and reflects a great deal of the legal analysis of why states comply with international rules without the enforceability of strong sanctions, is the robustness of the treaty itself. Modern treaties have various built-in systems that allow their parties to review a treaty's status through scientific mechanisms or effectiveness review systems, or that enable parties to learn and become familiarized with the problem the treaty is addressing. Combined with these review and learning mechanisms are additional built-in systems that allow treaty renegotiation and take on deeper commitments to tackling the problems they have been created to address.

A theory of effectiveness must also pay attention to the supporting components of the treaty which may not be binding on the parties directly but implementing these provisions enhances and enables parties to achieve the goals of the treaty. These include financing, national programmes, technology transfer, capacity-building, and even institutional parts of the treaty such as the treaty secretariat. Though measuring the implementation of these supporting components of the treaty from the viewpoint of behavioural change is again methodologically difficult, it is logical that these provisions do have an impact, and measuring the degree of implementation in terms of the number of programmes or the level of financing is also an important component of determining the effectiveness of a treaty. Though not a key legal requirement of treaty effectiveness, financing is nevertheless a crucial lesson that has been learned from treaty-making in the past.³⁵

International responses to new and emerging environmental challenges should therefore be crafted with an eye to their "effectiveness". As suggested by Chambers, a wider, rather than narrow, conception of effectiveness appears appropriate.

Similarly, while the first generation of international environmental law paid lip service to the clear link between human well-being and the environment (most notably in the repeated articulation of concepts such as the right to development and, more recently, sustainable development), increasingly international environmental law has become fragmented and disconnected, especially from other areas of international law. International environmental law is increasingly viewed as something separate from human rights law and international trade law, for example. But the reality is that the issues touched upon by all these bodies of law are very much interlinked. Failure to see these linkages has also undermined the effectiveness of international environmental law. As one scholar has noted:

Traditionally, international environmental protection and international economic law have been treated separately. The focus has been on controlling specific pollutants or conserving particular species as ends in themselves. In the new model, the focus is on ecosystems conservation, pollution prevention, and a precautionary approach, not only as environmental goods but as integral to sustainable development.

In the quest for environmentally sustainable development, the focus will likely be on considering environmental concerns at the front end of the industrializing process, so as to prevent pollution, minimize environmental degradation, and use resources more efficiently. This should mean increasing concern with making the entire production system environmentally sound. International environmental law will need to reflect this emphasis by focussing on performance standards to prevent pollution and minimize degradation, rather than on liability for damage, and on providing incentives to companies to use environmentally sound processes. Environment and trade issues will be increasingly joined.³⁶

The development of international law in the future will need to be developed mindful of these obvious but often forgotten interlinkages.

On a somewhat related note, the effectiveness of international environmental law in responding to new and emerging environmental challenges will also be partly dependent on us breaking out of the pernicious influence of the pervasive concept of state sovereignty in international law. Clearly to view international relations, international policy and international law as only shaped by the nation state is to deny the reality that a range of non-state actors now shape world affairs and responses to the major global environmental challenges of our times. There is some truth to the observation that we are witnessing the emergence of a “kaleidoscopic” international legal and political system.³⁷ That is to say:

At the same time as globalization and integration are increasing, there is increased fragmentation within States and pressures for decentralized decision making . . . While there is growing integration and fragmentation, the international system is also becoming kaleidoscopic. Shifting, *ad hoc* coalitions and associations, as well as individuals are becoming important actors . . . This stage in the evolution of the international system, with its emerging bottom-up empowerment, may be characterised as kaleidoscopic. It is informal, and the actors and coalitions constantly change.

These developments pose challenges for the international legal system and at the same time opportunities to strengthen and expand the foundations of international law. International law must operate in a new multi-layered system consisting of States, international institutions, private sector and nongovernmental organization networks, the wide range of formal transnational bodies, . . . and the new kaleidoscopic pattern of informal coalitions and individual initiatives.³⁸

However, international law (or international lawyers?) have been slow to realize the implications of these changes. While a “kaleidoscopic” world may be emerging, international lawyers still see the world through myopic “rose-coloured glasses”, regarding only the nation-state as the legitimate bearer of rights and duties under international law. The effectiveness of international responses to new and emerging global environmental challenges will thus also, in part, depend on how they are crafted to reflect this new “kaleidoscopic” world. Non-governmental organizations, transnational corporations and intergovernmental organizations must have a role to play in shaping and implementing responses to these issues. Likewise legal, regulatory and policy responses will need to take many forms beyond the usual command-and-control instruments typically recognized by nation-state-centric models, and incorporate the other institutions which comprise regulatory systems – regulated entities themselves, their industry associations, and third parties, both public interest institutions and commercial actors.³⁹

In broad terms all of these issues are related to the wider issue of “international environmental governance”. But what does the term “governance” mean in the context of existing and emerging environmental challenges, which for the most part are not confined to one location, one country or even one region of the world? To understand that, we need to understand what we mean by “international” or, perhaps more appropriately, “global governance”. As Rosenau has posited:

Does it refer to a central authority that can exercise control over far-flung situations on a global scale? Or is it limited to the exercise of authority in particular situations, such as environmental threats ... which may be global in scope and especially dire? Or does it connote the sum of all diverse efforts of communities at every level to move towards goals while preserving their coherence from one moment to the next?⁴⁰

In the context of global environmental challenges the notion of a central authority has dominated much of the debate on global environmental governance. Proposals have included an international environmental agency, an international environmental authority, a world environmental organization and an international environmental organization.⁴¹ While theoretically possible, a central authority with respect to global environmental issues looks unlikely to become reality.

Realistically governance, and in particular environmental governance, now and into the foreseeable future will be about the “diverse efforts of communities at every level”, as Rosenau has suggested. Thus, in considering the likely contours of the future of international law, we need to focus less on new laws or treaties, although these will still be significant, and

more on governance. We need to take a much more holistic approach, looking beyond law as a solution to international environmental problems. Importantly our focus on environmental governance should emphasize achieving “good” environmental governance. By that term we mean governance that can produce measurable and verifiable improvements in the state of the global environment.

The structure of this book

This book aims to explore some of these themes in more detail; to explore the future of international environmental law, not from the narrow perspective of what new treaties or new institutions may emerge, but from the broader perspective of international environmental law as but one part of the overall future of environmental governance. The book aims to contribute to debate on the future of international environmental law, the emerging issues it may need to tackle, and the relationship of those issues to development, human rights and other areas of international law.

The book is structured in three parts. Assessing the implementation experiences and the effectiveness of contemporary international environmental law offers us some valuable lessons on what we need to do to improve the effectiveness of international environmental law into the future. This will be important for assessing what needs to be done in terms of improving the effectiveness of existing law, and also the way in which international environmental law will be part of overall global environmental governance into the future.

Part I, the experience to date, continues with Chapter 2, in which Ann Powers highlights that, while climate change and pollution affect all states and peoples around the world, small island developing states are especially vulnerable. Developing countries, and small island developing states in particular, already face major challenges in dealing with pollution. But responses to existing environmental challenges such as pollution are complicated by the new and emerging challenges presented by climate change. No response to an environmental issue can be considered in isolation from other issues, and as this discussion highlights, the difficulties which small island developing states and other vulnerable coastal states face in managing and effectively implementing existing regional and global environmental agreements are only exacerbated by climate change concerns. Treaty congestion is clearly a real and pressing issue for such countries, even before one factors in the challenges now presented by climate change.

Moving beyond climate change, in Chapter 3, Susan Shearing examines the key challenges that affect the capacity of countries to develop and implement effective biodiversity conservation regimes. Her analysis of those challenges reveals the critical importance of addressing the key drivers for biodiversity loss through economic measures and institutional reform. There is a brief overview of the international legal framework for biodiversity conservation, focusing on the Convention for Biological Diversity (CBD) and associated policies and initiatives. The chapter goes on to argue that, notwithstanding the ongoing challenges for effective national implementation of the CBD obligations relating to conservation and sustainable use of biodiversity, a range of examples drawn from northern and southern nations illustrate the extent to which some key mechanisms identified in the CBD itself, including protected areas networks, economic tools and the sectoral mainstreaming of biodiversity issues, are playing an important role in facilitating national approaches to biodiversity conservation. The chapter thus illustrates that the effectiveness of the CBD goes beyond formal compliance with the letter of the convention; effectiveness depends very much on how those obligations are translated into concrete action on the ground.

Chapter 4 by Tullio Scovazzi paints a picture of one of the success stories in international environmental law – the legal instruments applying to the protection of the Mediterranean environment, the so-called Barcelona system of marine pollution treaties. The Barcelona system has acted as a model for the UNEP regional seas treaties and this chapter highlights why it is important for international legal regimes to be able to adapt over time as new challenges emerge.

This is followed by Chapter 5 where Donna Craig and Michael Jeffery explore the role of public participation and access to information in achieving sustainable development and good environmental governance. They describe the close link between sustainable development and transparency and access to information in fostering good environmental outcomes. Clearly good environmental governance at the national and international level involves many actors and many stakeholders, and in this chapter the authors explore existing legal principles and “best practice” approaches to environmental governance highlighting public participation, and its relationship with the ever-evolving goals and complexities of ecologically sustainable development. In that regard they highlight the particularly significant model presented by the Aarhus Convention.

In the last chapter in Part I, Gudmundur Alfredsson, continuing some of the themes examined in Chapter 5, goes on to explore the connections between human rights law and the protection of the environment, with a particular focus on the rights of indigenous peoples. This chapter focuses in particular on the rights to land and resources of indigenous peoples

and how these relate to environmental qualities; it covers the efforts at mainstreaming or incorporating human rights into other UN programme activities as well as human rights elements in international environmental instruments and how they are applied. A central question asked is whether environmental law can draw lessons from human rights law including, in particular, public access to and contributions to monitoring institutions. As this suggests, questions surrounding environmental governance are closely linked to developments in law and governance in other areas.

Part II of the book then goes on to explore the dynamics and challenges of a number of international legal regimes that are currently in transition. In Chapter 7, Michael Gerrard and Dionysia-Theodora Avgerinopoulou consider the challenges to environmental law posed by climate change, with a particular focus on the adaptation needs of both developed and developing countries. After considering the current status of international treaties relating to climate change, the chapter examines some of the key issues that need to be addressed beyond 2012. It also explores the interaction between international human rights provisions and climate change law. The authors argue in particular that climate change is becoming the public face of unsustainable development, and contend that new legal instruments will be required, but beyond that, developments in policy will also be important.

In Chapter 8, Timo Koivurova and Sébastien Duyck examine some key principles that could be used to sustainably address challenges in governing the Arctic Ocean. Significantly, as the first part of this chapter highlights, the environmental governance regime for the Arctic already clearly recognizes the legitimate expectations of and a role in decision-making for non-state actors, and in particular the indigenous peoples of the Arctic. As the authors make clear, future discussions in relation to the Arctic will focus on governance rather than on new treaties.

In Chapter 9, Rosemary Rayfuse examines the relevance of the Grotian legacy of freedom of the high seas for the future of international environmental law relating to the marine environment. She argues that the history of the law of the sea has been one of oscillation between freedom and restriction. Today, she argues, freedom of the high seas is not unlimited and an ever-increasing range of restrictions on this freedom now exists. Rayfuse goes on to consider what might replace the Grotian legacy in the future.

Part III examines a number of new and emerging issues for international environmental law to address. In Chapter 10 the editors of this book, David Leary and Balakrishna Pisupati, consider the implications of the emergence of nanotechnology. This chapter focuses on the emerging global debate on the need for new regulation to address perceived

environmental and human health risks associated with nanotechnology. This analysis highlights that the response of policymakers, regulators and the legal system is very much a work in process. The chapter argues that it appears increasingly obvious that international responses to nanotechnology relate not so much to law, per se, but perhaps falls under the broader idea of international governance for nanotechnology. The evolution of this governance, perhaps eventually leading to new laws and treaties, will need to be driven by increased scientific understanding and a precautionary and adaptive framework of risk management.

This is followed, in Chapter 11, by Richard Ottinger and Victor Tafur's examination of the risks and opportunities posed by the emergence of biofuels as an alternative fuel source. The chapter focuses on emerging responses such as the development of appropriate standards to assure the sustainability of biofuel production, again highlighting the potential of governance mechanisms beyond new laws and treaties. New standards for sustainable biofuels appear initially to be most significant, pending development of any future treaty or other "hard law" responses.

In Chapter 12, Michele Garfinkel and Robert Friedman explore the emergence of yet another novel issue for international policymakers to consider, namely, synthetic biology. In this chapter the authors first describe the science and engineering of synthetic biology and synthetic genomics and then go on to discuss emerging societal concerns about the technologies. There has so far been little consideration of the implications of this issue for international environmental law, so this chapter will no doubt serve as the starting point for a debate on a new and as yet unconsidered challenge for international environmental law and policy. As the starting point for a new and significant debate, the chapter highlights that synthetic biology and synthetic genomics introduce novel problems for those concerned about the governance of biotechnologies generally. Responding to those challenges is both a legal and a broader policy and governance challenge.

In the concluding chapter the editors draw together the main conclusions from all the contributions and consider their implications for the future of international environmental law.

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United Nations University, 53-70, Jingumae 5-chome,
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Tel: +81-3-5467-1212 Fax: +81-3-3406-7345
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Edited by David Leary and Balakrishna Pisupati

Contributors:

David Leary
Balakrishna Pissupati
Gudmundur Alfredsson
Dionysia-Theodora
Avgerinopoulou
Donna Craig
Sébastien Duyck
Robert M. Friedman
Michele S. Garfinkel
Michael B. Gerrard
Michael Jeffery
Timo Koivurova
Richard L. Ottinger
Ann Powers
Rosemary Rayfuse
Tullio Scovazzi
Susan Shearing
Victor M. Tafur

This book explores the future of international environmental law in a world of ever worsening environmental crises. It examines the success stories and failures of international environmental law and argues that future responses to global environmental crises will be more about good environmental governance than just more treaties and laws. Environmental governance will need to accommodate the needs and aspirations of peoples from developed and developing countries alike and will have to be based on decisions and actions by a vast range of actors and stakeholders—not just the nation-state that has traditionally dominated environmental diplomacy. This also suggests a need to be cognizant of the close links to other areas of international law, including human rights.

The Future of International Environmental Law tackles the major environmental challenges of our times including climate change, biodiversity loss, and pollution and overfishing of the oceans. It examines what we can learn from the implementation of existing international environmental laws over the past few decades. It also considers a range of emerging issues such as the management of the environmental challenges faced by the Arctic, nanotechnology, biofuels and synthetic genomics.

David Leary is a Senior Research Fellow in the Faculty of Law, University of New South Wales, Sydney, Australia.

Balakrishna Pisupati is Coordinator of the Biodiversity, Land Law and Governance Unit, and United Nations Environment Programme (UNEP) Focal Point for the Convention on Biological Diversity and the International Year of Biodiversity in the Division of Environmental Law and Conventions, UNEP, Nairobi, Kenya.

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